जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

05th March 2025

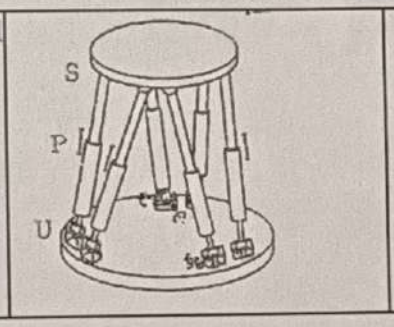
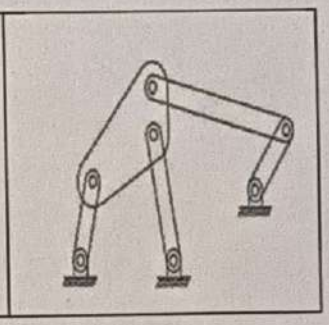
Roll No. | | | | | | | | | | | | |

MID SEMESTEREXAMINATIONS

Programme	:	B.Tech CSE / B.Tech CSE (Cyber Security)	Semester	:	VI
Course Name	:	Robotics and AI	Course Code	:	BECCS3C009 / BECSE3C009
Duration	:	**1.5 Hours**	Credits	:	**03**
Maximum Marks	:	45	Minimum Marks	:	

After pursuing this course, the student will be able to:	
CO 01	Learn the fundamental principles, components, and applications of robotics and AI.
CO 02	Understand the designing methodology of integrating actuators, sensors, and control mechanisms in robotic systems.
CO 03	Explore AI techniques such as NLP, computer vision, and reinforcement learning for robotic systems
CO 04	Investigate tools like ROS and Gazebo for robot simulation and assess IoT integration in robotics.
CO 05	Explore robotic process automation (RPA) tools and their applications in various domains.

Part A Very Short Answer Type Logical Questions of 02 marks related to different COs		**Total Marks** **5X2= 10**	
S. No.	**Statement**	**Levels**	**CO's**
A1	Draw a basic robotic system design of your choice and highlight the key components from the same.	3	1,2
A2	Define actuators? Enlist the types of actuators.	1	2
A3	Explain intelligent agents and list their types.	2	3
A4	What does it mean by Rational Agent? Enlist its properties.	2	1
A5	Discuss the types of Batteries (Power Supply) used in robots.	2	1,2
PART B Short Answer Type Analytical Questions of 3 marks related to different COs **(Attempt any five questions)**		**Total Marks** **5X3=15**	
B1	Outline the design principles for robot user interfaces.	3	2
B2	List Asimov's Laws of Robotics and discuss their implications.	4	1
B3	Identify and highlight the artificial intelligence techniques used in robotics and provide examples.	4	3
B4	Explain utility-based agents with a diagram and an example.	3	2,3
B5	Describe various types of communication systems for robots with its suitable example.	3	2
B6	What does it mean by following terms with respect to Degree of Freedom 1. Negative DOF 2. Zero DOF 3. Positive DOF	2	2

Part C Long Answer Type Descriptive Questions of 10marks related to different COs **(Attempt any Two questions)**		Total Marks 2X10=20	
C1	Discuss the types of agents used in robotics, providing examples for each type and supporting your answer with diagrams.	4	2,3
C2	Explain Various types of motions in robotics. Also, discuss the concept of Envelope with respect to robotics. Various types of joints in robotics along with their Degree of Freedom.	3	1,2
C3	Define the Degree of Freedom (DoF) of a Robot, Calculate the DoF for following: **1**  **2** 	5	2